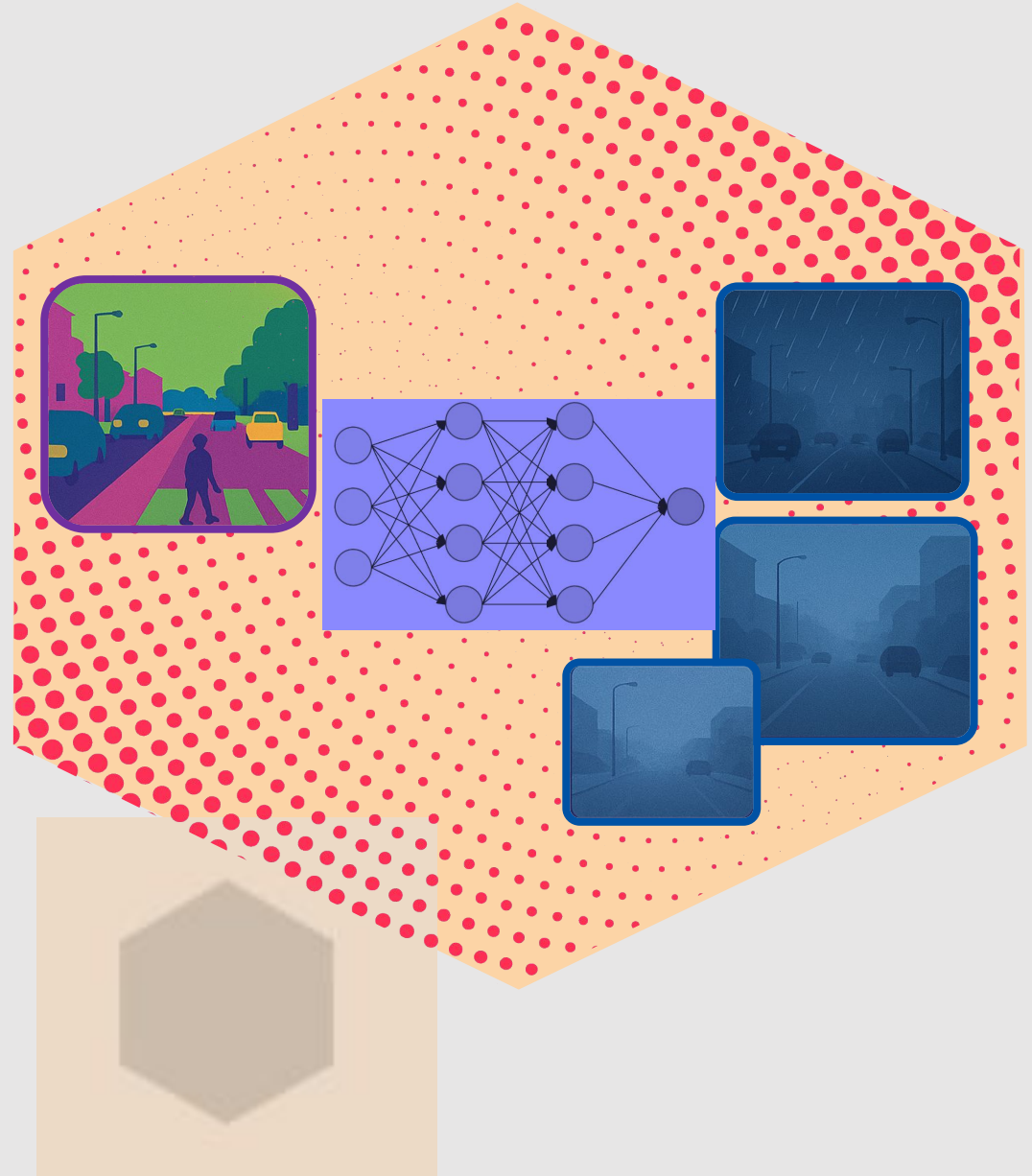
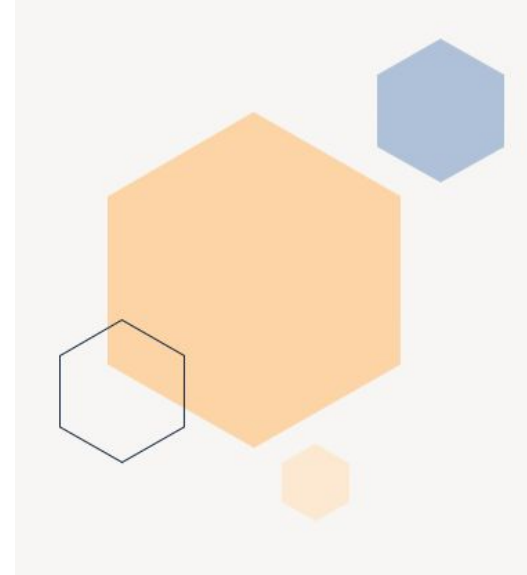


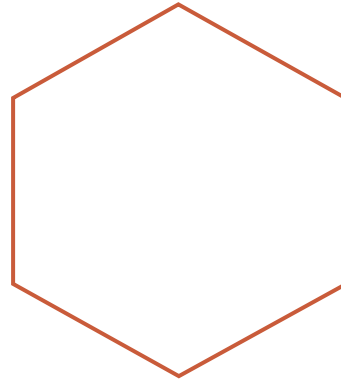
Semantic-Guided GAN Framework for Generating Adverse Weather Conditions

Samar Fathalah



- **Introduction**
- **Literature Review**
- **Methodology**
- **Evaluation**
- **Results**
- **Conclusion & future work**





INTRODUCTION



- Object detection is **central** to autonomous driving systems.
- However, most models are trained on datasets captured in clear weather, limiting their real-world performance.
- Commonly used datasets for urban object detection include:



Cityscapes dataset [1]



KITTI dataset [2]

- Real-world adverse weather conditions (e.g., rain, snow and fog) are often underrepresented in existing datasets.
- Existing datasets **lack aligned** adverse-weather, annotation, and clear-weather **pairs**.
- There's a growing demand for realistic synthetic images that accurately depict complex weather effects like **fog density** and **rain reflections**.

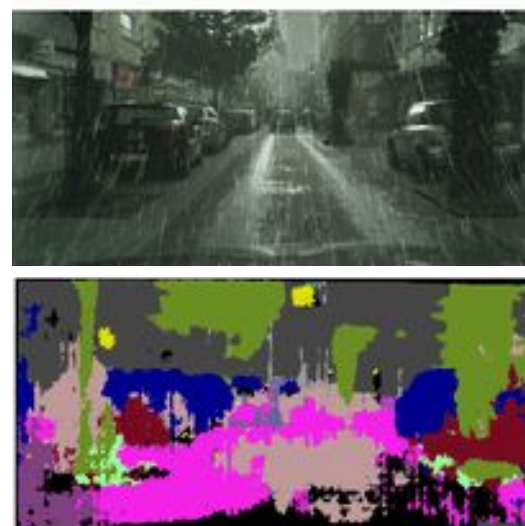
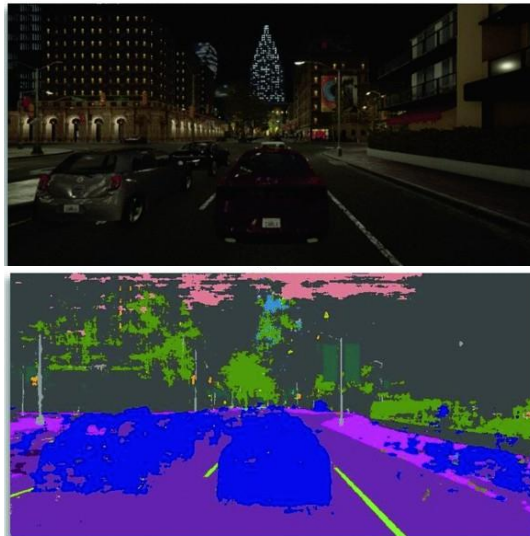


Snowy scene — ACDC dataset [3]



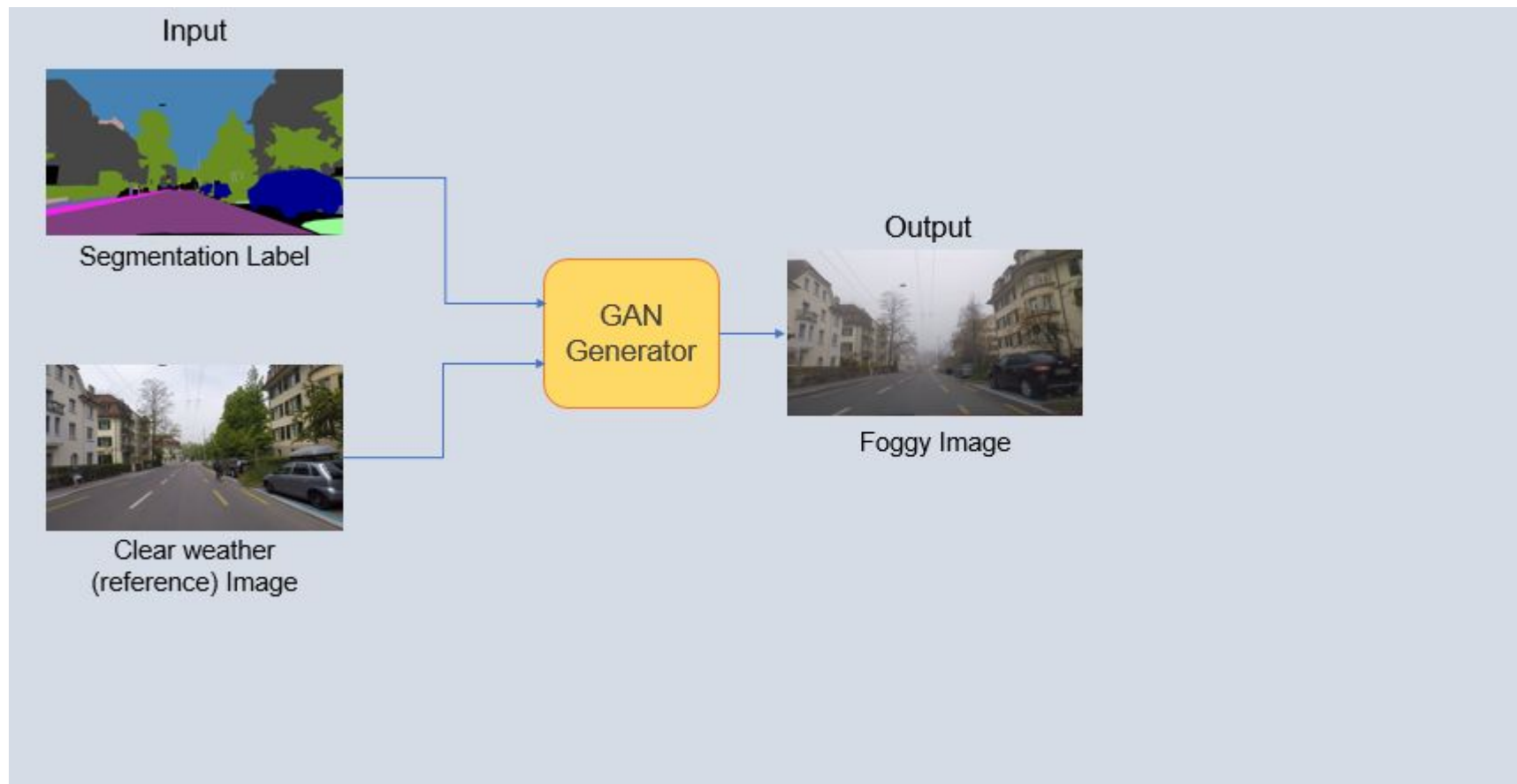
Foggy scene — ACDC dataset [3]

- Existing methods used to generate synthetic images of adverse weather conditions struggle to preserve *spatial* and *semantic* consistency.
- Spatial consistency and object boundaries often get distorted in fog, rain, or low light.
- Using semantic maps provide structured scene layout, ensuring that key objects (e.g., roads, vehicles, pedestrians) are preserved.



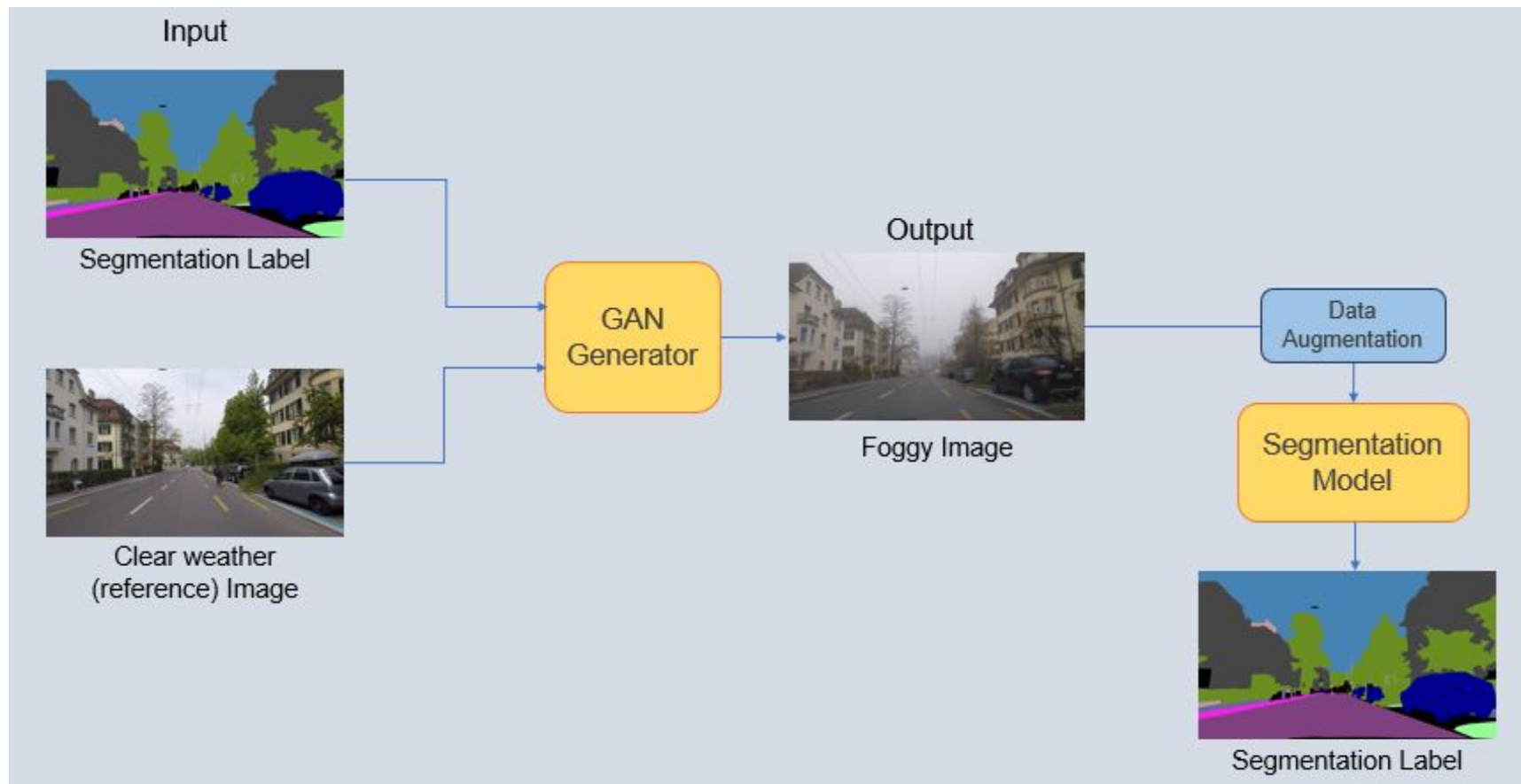
Example of underperforming Segmentation Models on Images with Adverse Weather Conditions

- Develop a model that generates foggy images *from clear-weather inputs*.
- The generation is *guided by semantic information*.
- Approach: Generative Adversarial Networks (GANs).

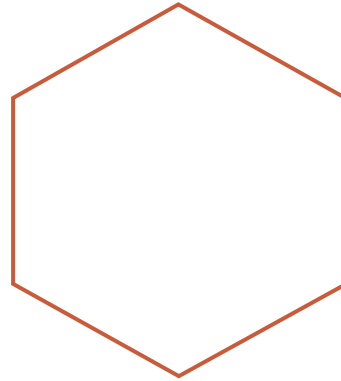


Proposed Framework

- Develop a model that generates foggy images *from clear-weather inputs*.
- The generation is *guided by semantic information*.
- Approach: Generative Adversarial Networks (GANs).



Proposed Framework

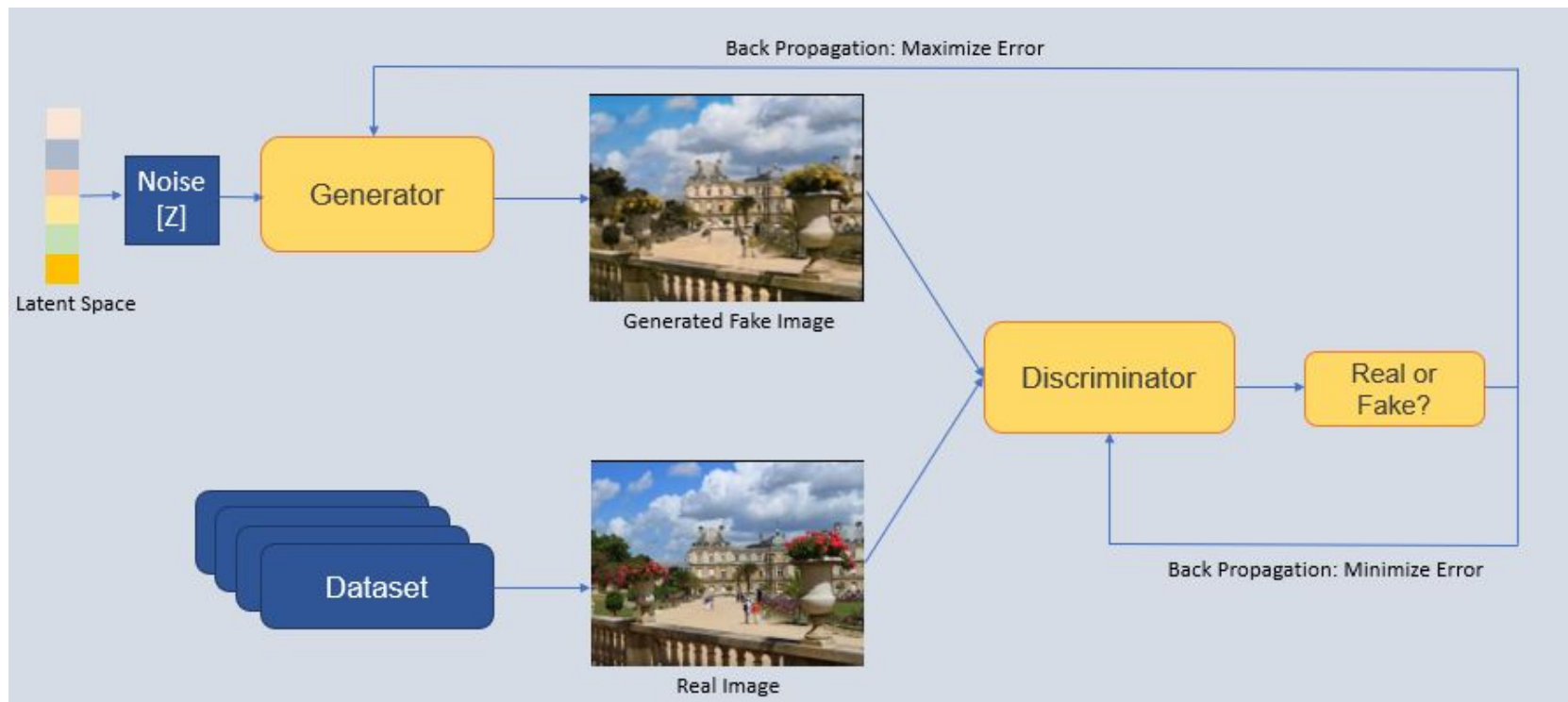


LITERATURE REVIEW



Literature Review: GAN framework

- Generative **A**dversarial **N**etworks introduced by Ian Goodfellow et al. in 2014 [4].
- **GAN training** is a two-player adversarial game between:
 - Generator (G): Learns to create realistic images to “fool” the discriminator.
 - Discriminator (D): Learns to distinguish between real and generated (fake) images.
- **Goal:** Train the generator to produce outputs that are indistinguishable from real data.



Visualization of GAN model

- Image-to-Image translation task: A computer vision task that learns to **map** an input image **from one domain** to a corresponding image in **another domain** [4].
- Semantic Image Synthesis: A task of generating photorealistic images from semantic segmentation maps [6].
- Typically solved using deep learning, especially Generative Adversarial Networks (GANs).



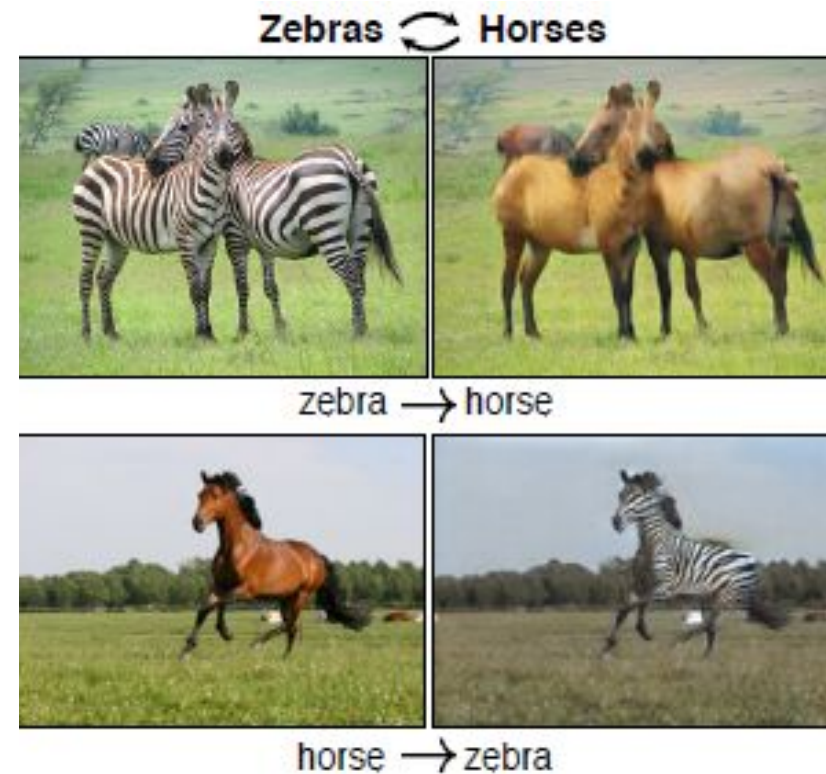
I2I — Grayscale to Color [4]



SIS — Label to Image [6]

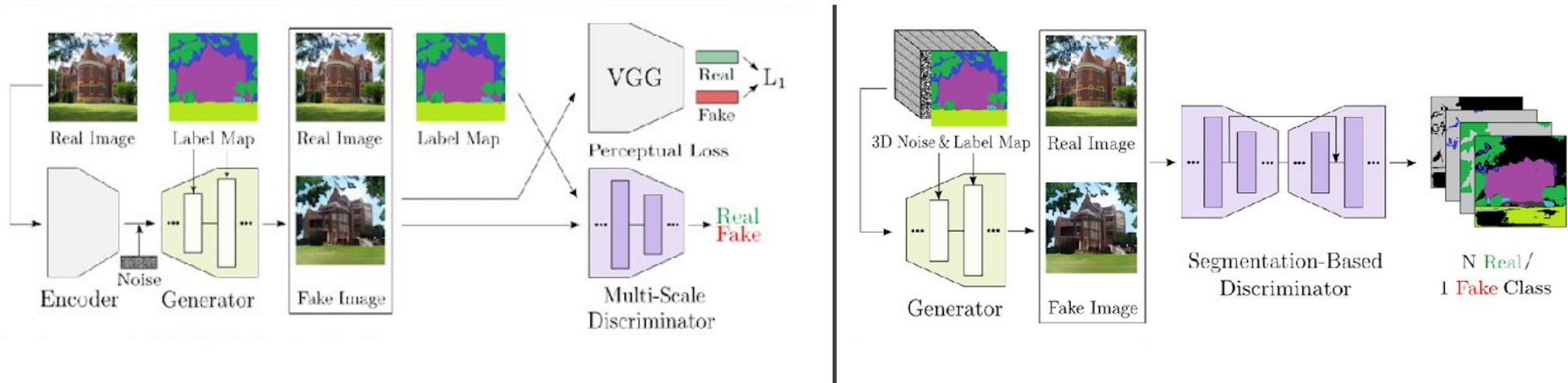
- Designed for **unpaired** image-to-image translation, where direct input-output image pairs are not available [5].
- Learns to translate between two visual domains using **cycle consistency**.
- Cycle Consistency Loss ensures that translating to the other domain and back yields the original image.

$$F(G(x)) \approx x, \text{ and } G(F(y)) \approx y$$



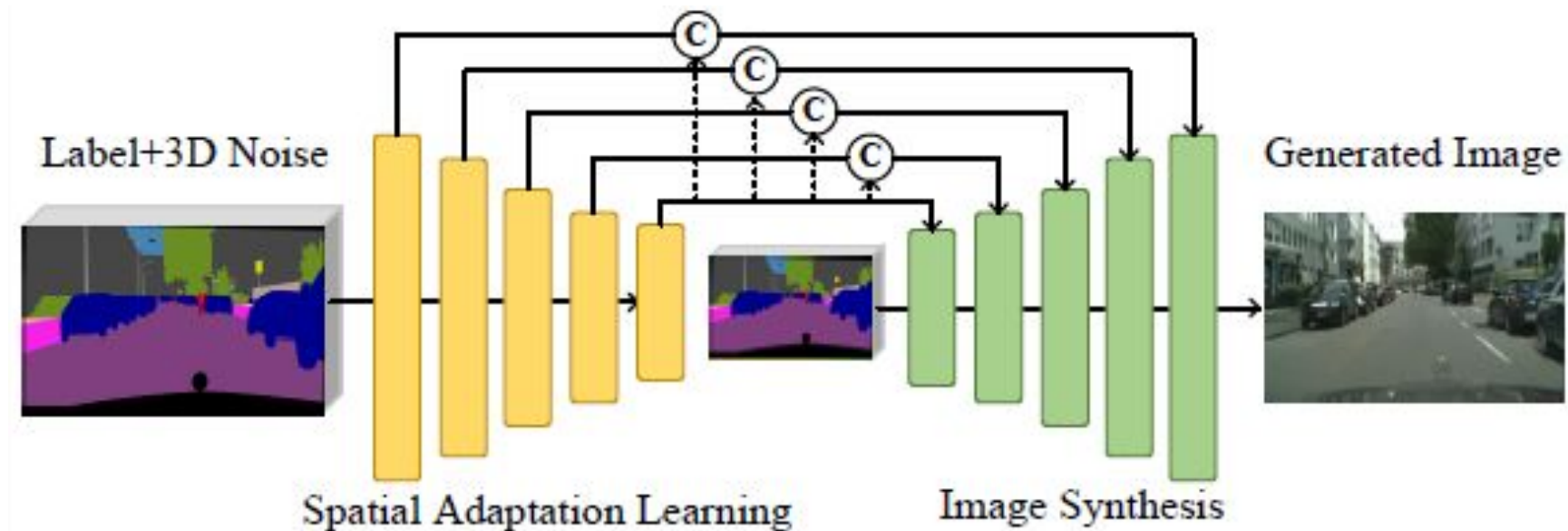
Cycle Consistency Example [5]

- Extends on SPADE with **object-aware discriminators** to focus on individual semantic classes (e.g., cars, buildings).
- Segmentation-Based Discriminator
 - Treats semantic image synthesis as a per-pixel classification task.
 - Each pixel is classified as one of the semantic classes or a “fake” class.



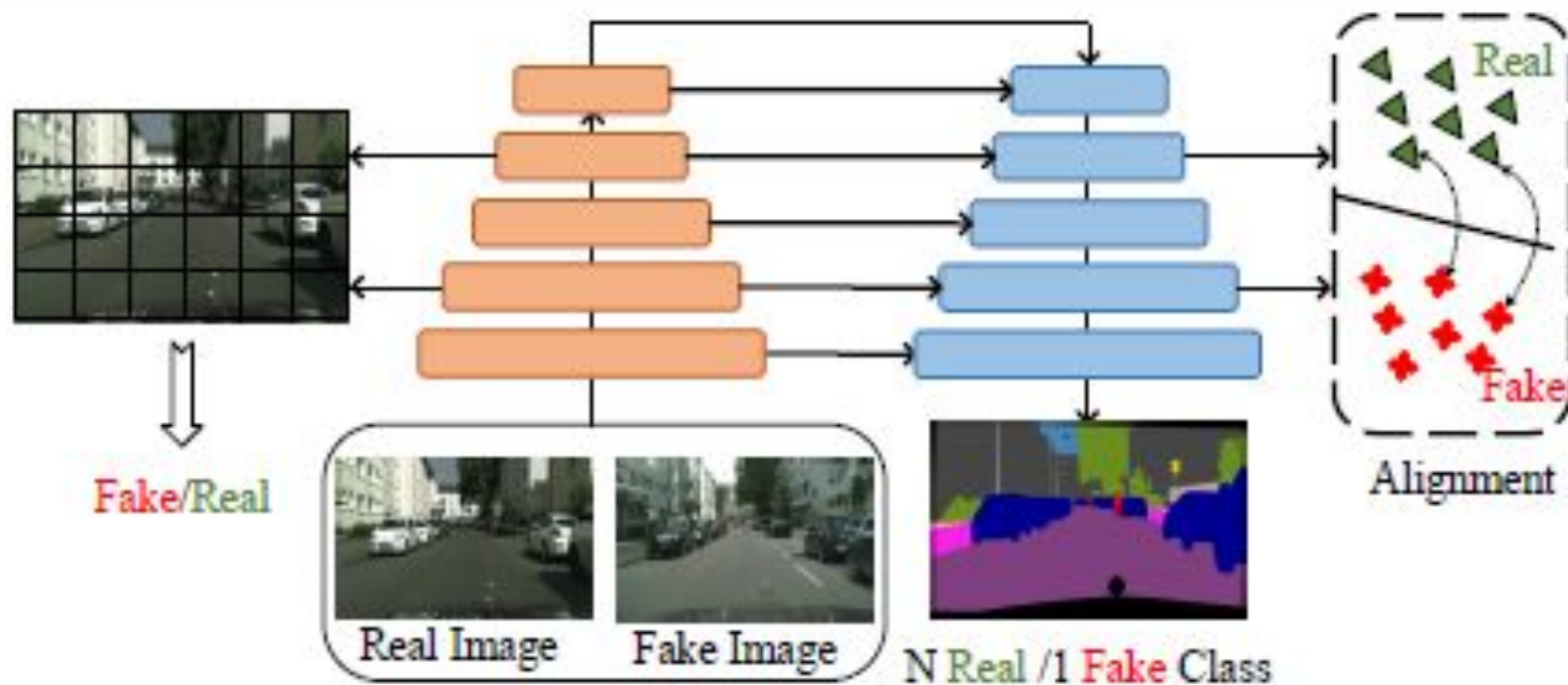
SPADE vs OASIS [8]

- **Dual Pyramid Generator:**
- Introduces two interconnected pyramids:
 - **Conditioning Pyramid:** Learns spatially-adaptive normalization (SPADE) parameters from full-resolution semantic maps.
 - **Image Pyramid:** Generates images through upsampling layers.

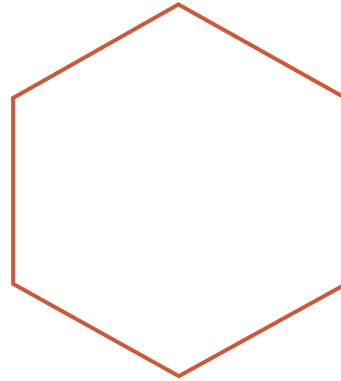


DP GAN Generator Architecture [9]

- **Scale-Enhanced Discriminator:**
- Incorporates **multi-level** supervision:
 - **Pixel-level:** Ensures fine-grained detail accuracy.
 - **Patch-level:** Maintains local consistency.
 - **Feature-level:** Preserves global structural coherence.



DP GAN Discriminator Architecture [9]



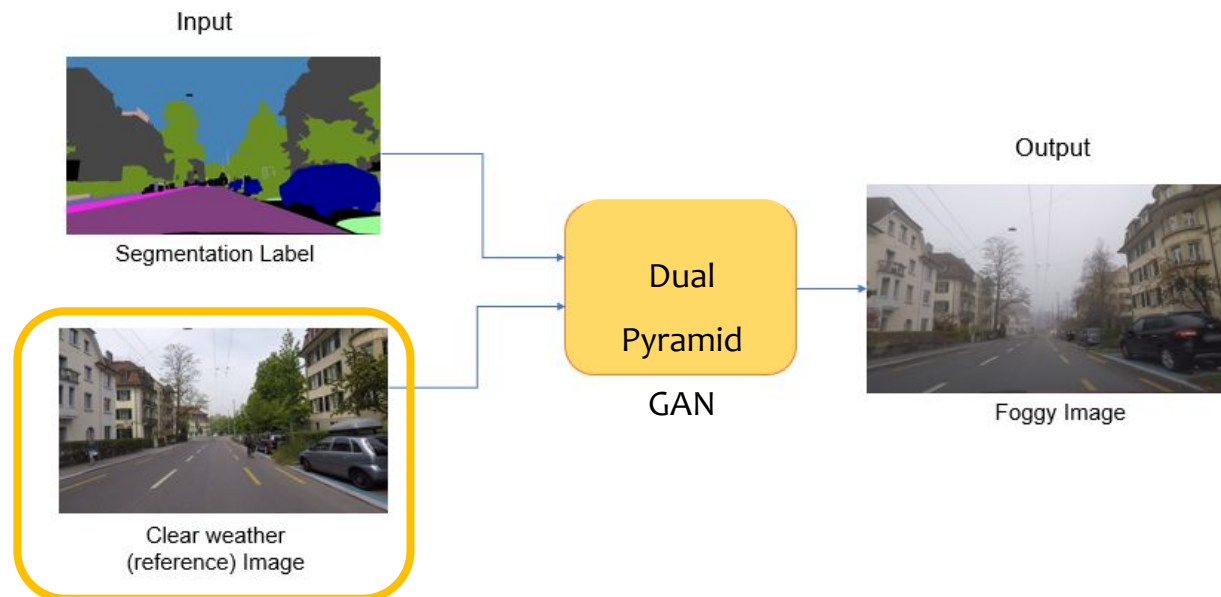
METHODOLOGY



Methodology: Clear Weather Incorporation

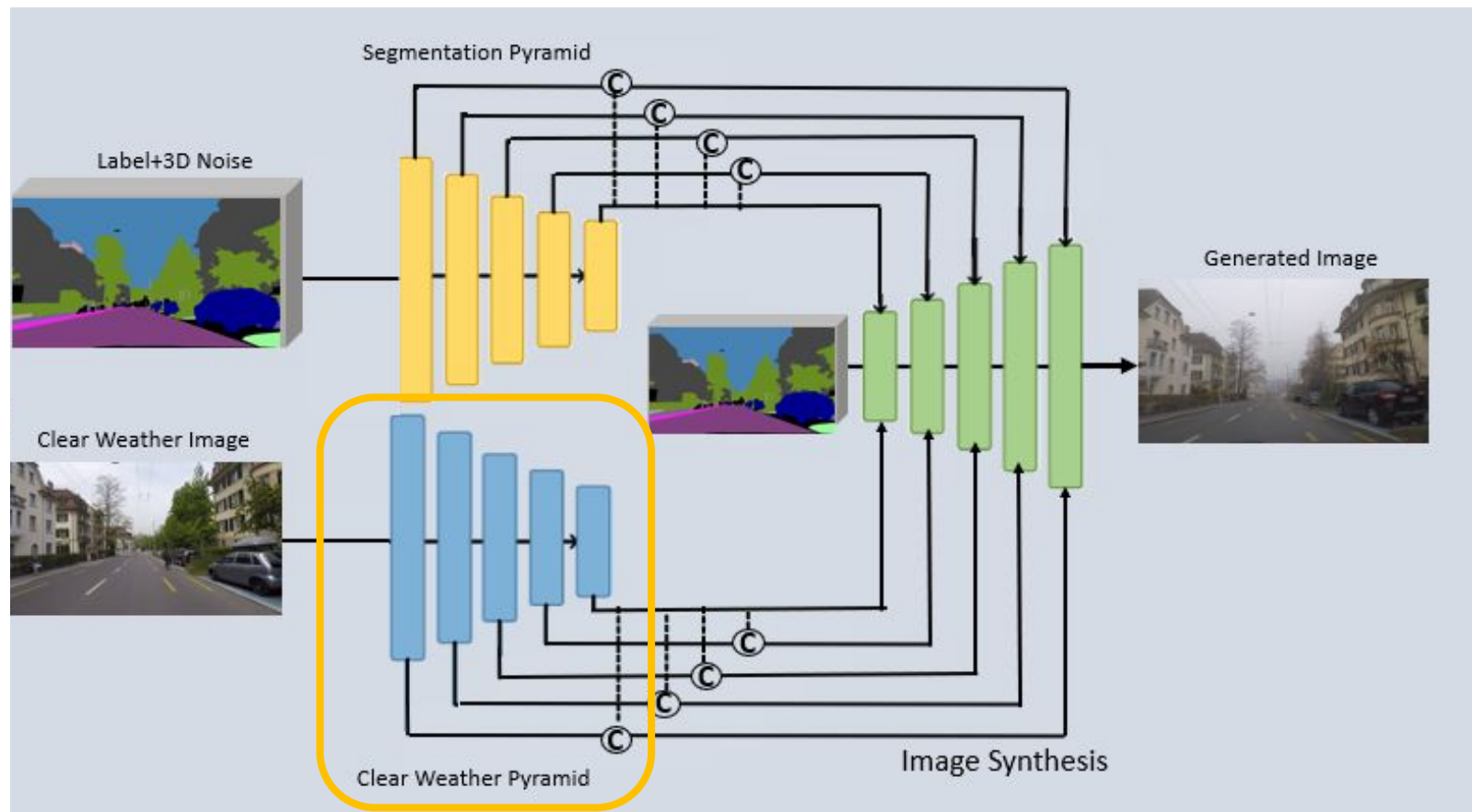


- Introduced the corresponding **clear-weather** image as an additional input.
- Inspired by Image-to-Image Translation Models:
 - Builds on ideas from models like **Pix2Pix** [10] and **CycleGAN** [5] that learn mappings between visual domains.
 - Enables the GAN to model how specific features change under adverse conditions.
- **Multi-Modal Conditioning:**
 - The model now uses **semantic maps** for structure and **clear images** for texture/context.

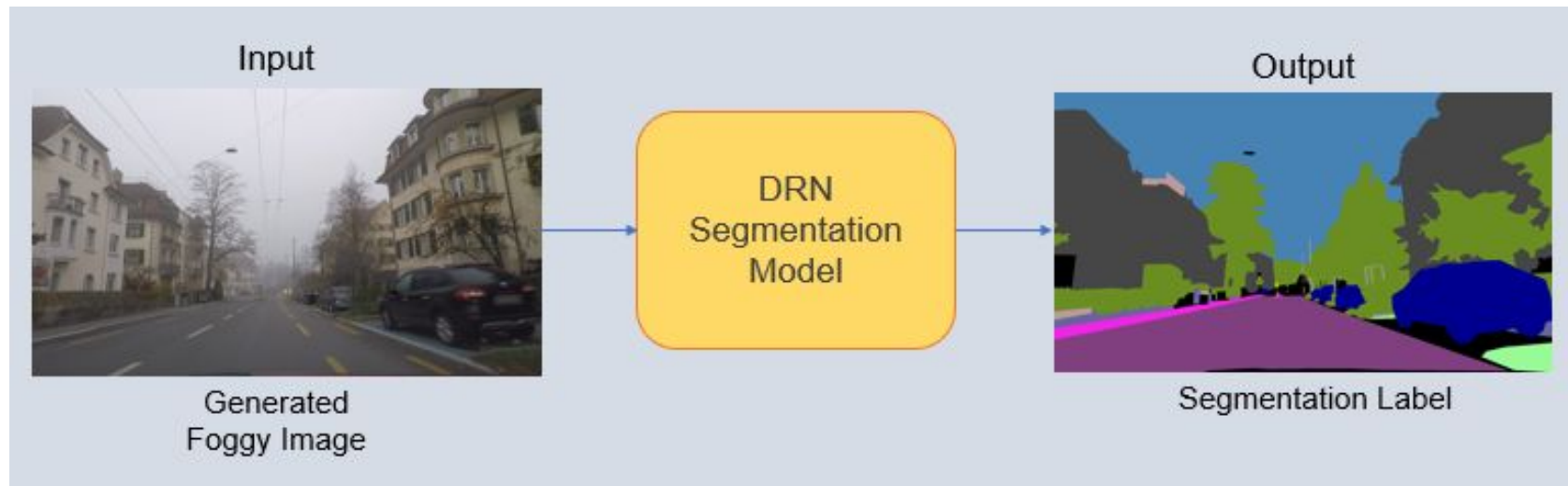


- Selected **DP-GAN** due to its higher performance in both *realism* and *semantic* consistency [9].
- Key Strengths of DP-GAN:
 - Maintains *global structure* and *local detail* — critical for urban scenes.
 - Demonstrated *robustness* on small and imbalanced datasets, unlike heavier models prone to overfitting.
 - Utilizes SPADE-based semantic conditioning and dual-path generation, enhancing fog representation.
 - Outperforms OASIS by improving not just the discriminator but also the generation process itself.

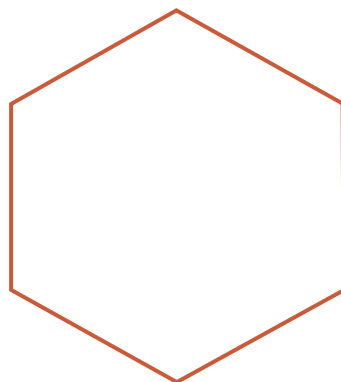
- Modified Generator Architecture:
 - **Pyramid 1:** Encodes *semantic segmentation maps*, capturing structural and spatial layout.
 - **Pyramid 2:** Performs the *multi-scale decoding and synthesis*, generating the adverse weather image.
 - **Pyramid 3 (new):** Encodes *the clear-weather image*, extracting multi-resolution visual features.



- Evaluating Semantic Consistency with **DRN** [11].
- Training **Dilated Residual Network (DRN)**.
- **Data Augmentation** using generated foggy images .
- To assess how well generated images preserve meaningful scene content.
- Serves as an indirect yet rigorous test of how well our generated images retain label-consistent structure.



Segmentation Model Framework



EVALUATION



- **ACDC** (**A**dverse **C**onditions **D**ataset with **C**orrespondences)[3].
- Contains **4,006** real-world images captured in European cities from vehicle-mounted cameras.
- Organized into four weather condition subsets: **Snow, Fog, Rain, and Night** (~1,000 images per subset).
- Images are labeled with dense pixel-wise annotations across **19 semantic classes**, aligned with **Cityscapes** taxonomy.



Weather Condition Samples from ACDC

- **Fog subset** was used for training, experiments suggested being the most challenging in for object detection tasks.



(a) Foggy Weather Image



(b) Foggy Annotation Image



(c) Paired Clear Weather Image



(d) Clear Weather Annotation Image

ACDC Fog Subset Sample

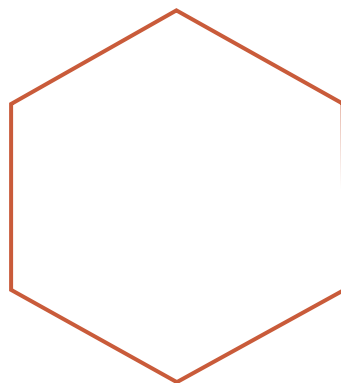
- Two Training Strategies.
 - Strategy 1: **Unpaired Image-to-Image** Translation using **CycleGAN** without semantic maps.
 - Strategy 2: **Semantic Image Synthesis** using **OASIS** and **DP-GAN**, which are explicitly conditioned on **segmentation maps**.
- Trained on **ACDC fog** domain.

- **CycleGAN:** Trained to translate from **clear to foggy** images **without semantic maps**.
- **OASIS, DP-GAN: Trained** to generate foggy images from **segmentation maps only**.
- Input images were resized to 512x256.
- Identical preprocessing pipeline and dataset split for fair comparison.
- No clear-weather images were used to test generation quality.
- The original models were **trained** without any architectural modifications.

- **3P-WeatherGAN** (Three *P*yramid *WeatherGAN*) proposed methodology.
- The training set included **400 fog-domain** images from ACDC, corresponding clear-weather images, and segmentation maps
- Training experiments setup:
 - **500 epochs**
 - **Batch size = 2**
 - **Single NVIDIA RTX A5000 GPU (24 GB)**
- Training used a combination of **VGG perceptual loss** and a **feature alignment**.
- The **weak pairing** property of ACDC ensures segmentation maps remain consistent.
- **50 validation** clear-weather images with exact paired fog-domain counterparts and annotations to simulate a **real-world deployment scenario**.

- Evaluate whether *synthetic foggy images* contribute positively to real-world segmentation performance.
- Trained **DRN-C-26** on *augmented data* of real and GAN-generated foggy images from ACDC dataset.
- Used standard **19-class label set** (Cityscapes-compatible).
- Training experiments setup:
 - **100 epochs**
 - **Batch size = 5**
 - **Learning rate decay at epoch 50**
 - **Single NVIDIA RTX A5000 GPU (24 GB)**
- Data augmentation included random rotation ($\pm 10^\circ$) and random scaling ($\times 2$).

- **Visual realism** of generated images:
 - Used **Fréchet Inception Distance (FID)** to assess generated image quality [12].
 - **FID** measures the distance between the feature distributions of real and generated images, using activations extracted from a pretrained Inception-v3 network.
 - **Low FID $\Rightarrow$ better realism.**
- **Semantic correctness** with respect to **input segmentation maps**:
 - Measured using **mean Intersection-over-Union (mIoU)** [13] on output of downstream task i.e. **DRN model**.
 - **Higher mIoU $\Rightarrow$ better semantic consistency.**



RESULTS



- Evaluated **pretrained model** “*nvidia/segformer-b5-finetuned-cityscapes-1024-1024*” across **ACDC** weather domains.
- Fog domain showed the **lowest mIoU**, indicating strong degradation in segmentation accuracy.
- Suggests fog is the **most challenging condition** for object detection tasks.

Training Set Adverse Weather Condition	mIoU
<i>Fog</i>	<i>41.2</i>
Snow	48.9
Rain	56.1

Table: Quantitative Comparison — mIoU results

- Evaluation of baseline methods versus our proposed method based on **Fréchet Inception Distance (FID)** — *lower is better*.
- Scores were calculated on ACDC Fog Validation Set (100 Images).

Method	FID
OASIS	122.10
CycleGAN	114.03
3P-WeatherGAN (Ours)	98.01

Table: Quantitative Comparison — FID results



Results: Visual Comparison



Original Image — ACDC

OASIS

CycleGAN

Ours



Table: Qualitative Comparison

Original Clear Image — ACDC

Ours



Table: Qualitative Comparision

Results: Downstream Task



- Evaluation of **segmentation task** on images generated by CycleGAN (**no segmentation input**) versus our proposed method (**segmentation-guided**) based on mean Intersection-over-Union (**mIoU**) — **higher is better**.
- The segmentation model was trained on **varying ratios** of real foggy images and GAN-generated images.
- Scores were calculated on ACDC Fog Validation Set with matching annotations (50 Images).

Training Dataset		mIoU
Real Fog Only		36.3
Real Fog + 200 synthetic images	CycleGAN	38.12
	Ours	41.42

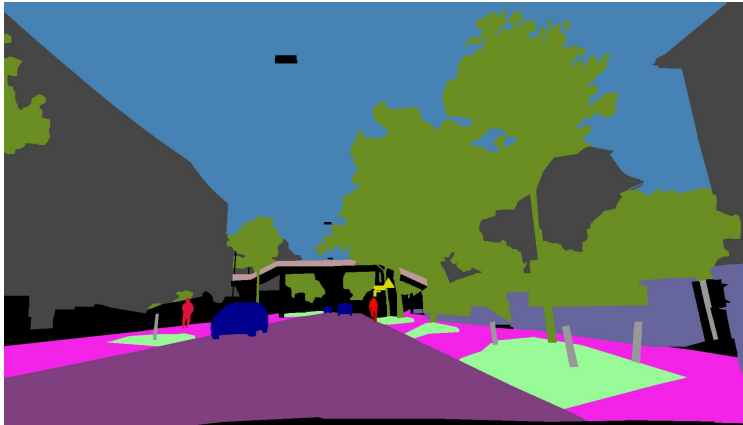
Table: Quantitative Comparison — mIoU results



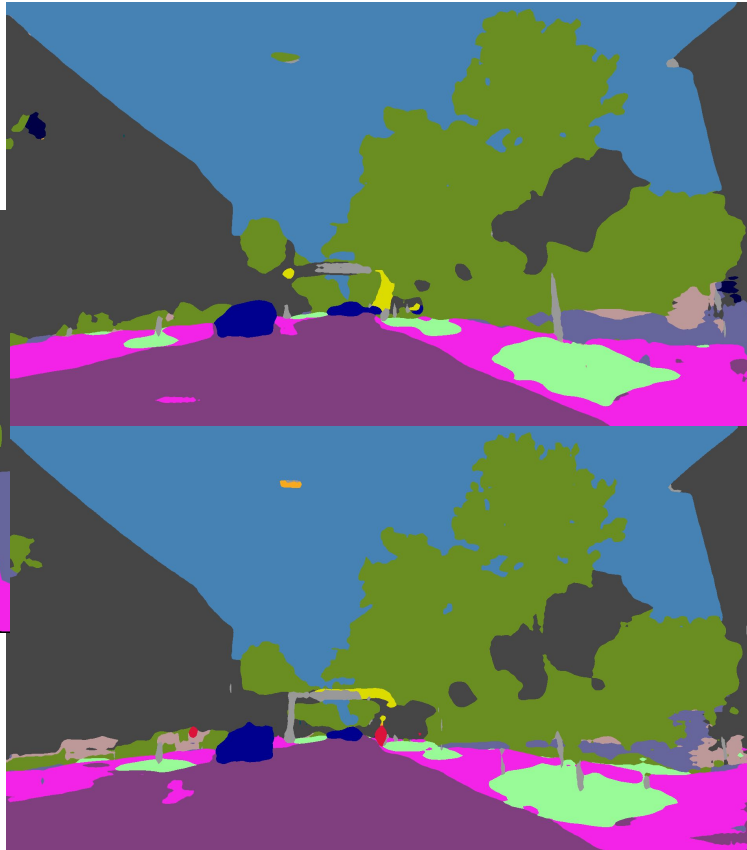
Results: Segmentation Visual Comparison



Original Label — ACDC

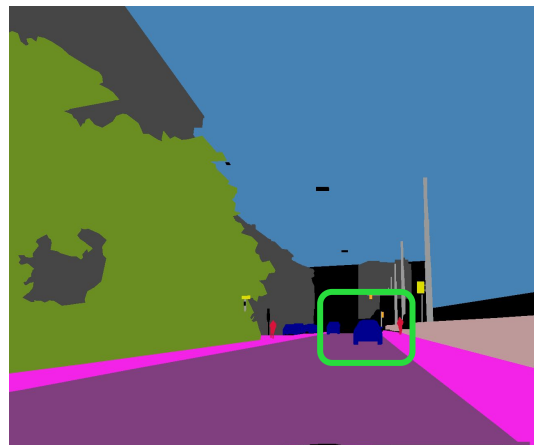
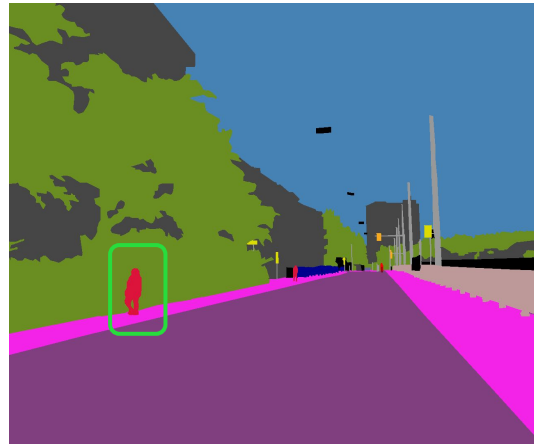
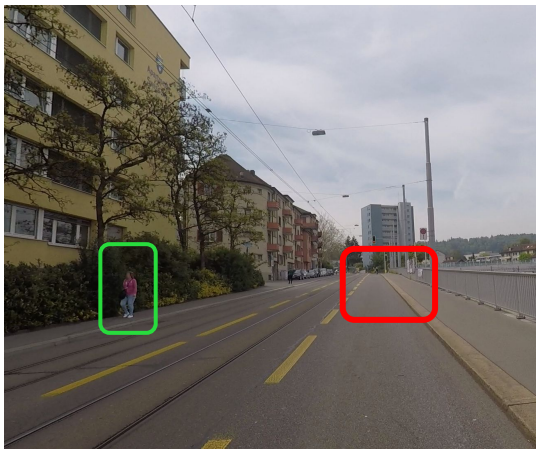


Real Only



Real + 200
Synthetic Images

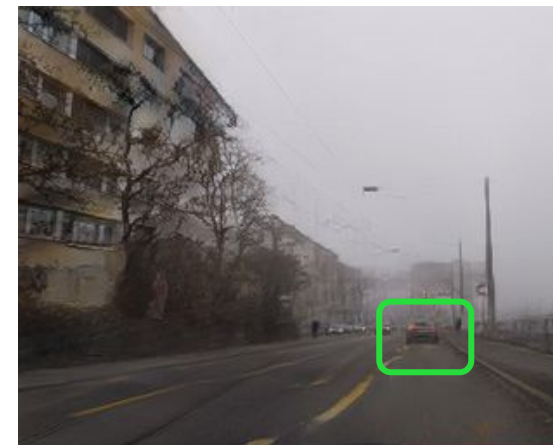
- Our method achieved **more realistic** outputs (**lower FID**).
- In the downstream segmentation task, our method initially **outperformed** other methods.
- Our method offers **scene-level control**.
- By leveraging both a segmentation map and a real image, our approach enables **explicit scene editing** — such **as inserting or removing objects** via the label map.



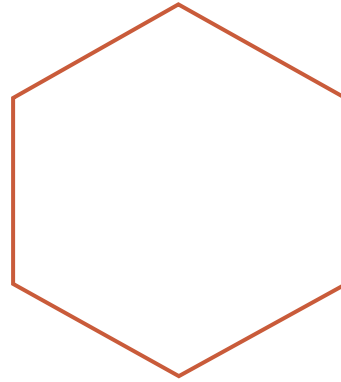
Input



Output 1



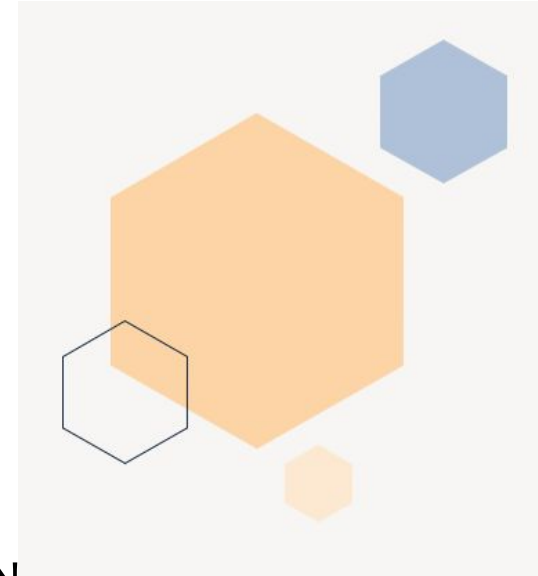
Output 2



CONCLUSION



- Proposed **3P-WeatherGAN** a *semantic-guided image synthesis framework* for generating images with adverse weather conditions.
- Our methodology achieved the **best visual realism**.
- **Outperformed other** unpaired methods like CycleGAN in downstream segmentation experiments.
- Our approach offers **greater semantic control**, allowing for scene editing via label maps — a key advantage over unpaired models.



Future Work

- Extend to ***other weather conditions*** like rain, snow, or compound scenarios to test generalization.
- Using ***more robust segmentation*** model.
- Incorporate ***explicit weather cues*** (e.g., depth, humidity, temperature) for richer context.
- Explore ***weather-aware loss functions*** to focus learning on affected regions.
- Investigate ***advanced generative models*** (e.g., diffusion models, Visual Language Models) for more control and diversity.





**THANK YOU
Questions?**

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- [3] Sakaridis, C., Dai, D., & Van Gool, L. (2021). *ACDC: The adverse conditions dataset with correspondences for semantic driving scene understanding*. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 10765-10775).
- [4] Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., ... & Bengio, Y. (2020). *Generative adversarial networks*. Communications of the ACM, 63(11), 139-144.
- [5] Zhu, J. Y., Park, T., Isola, P., & Efros, A. A. (2017). *Unpaired image-to-image translation using cycle-consistent adversarial networks*. In Proceedings of the IEEE international conference on computer vision (pp. 2223-2232).
- [6] Schönfeld, E., Sushko, V., Zhang, D., Gall, J., Schiele, B., & Khoreva, A. (2021). *You only need adversarial supervision for semantic image synthesis*. In Proceedings of the 9th International Conference on Learning Representations (ICLR).

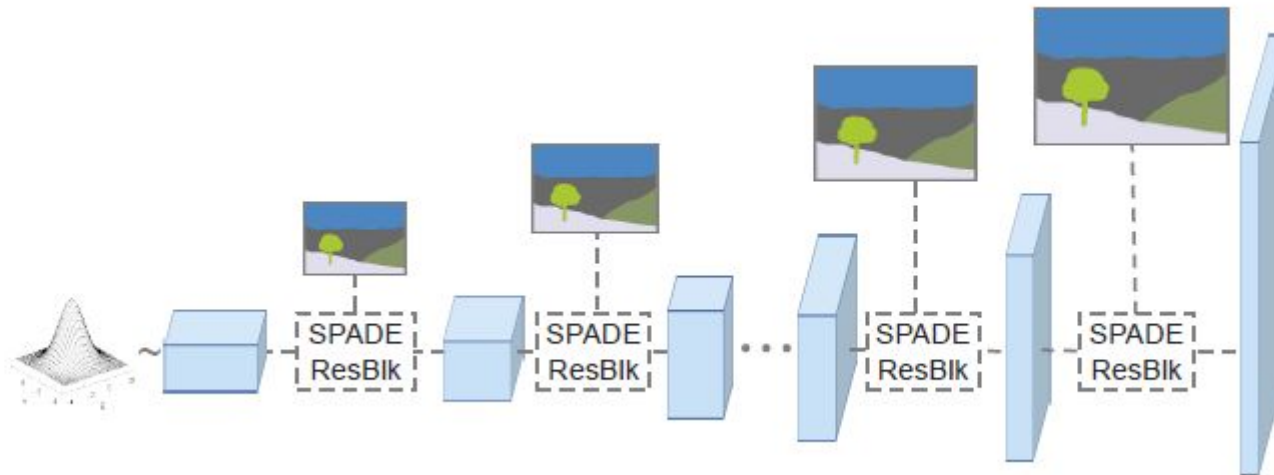


- [7] Park, T., Liu, M. Y., Wang, T. C., & Zhu, J. Y. (2019). *Semantic image synthesis with spatially-adaptive normalization*. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 2337-2346).
- [8] Sushko, V., Schönfeld, E., Zhang, D., Gall, J., Schiele, B., & Khoreva, A. (2022). *OASIS: only adversarial supervision for semantic image synthesis*. International Journal of Computer Vision, 130(12), 2903-2923.
- [9] Li, S., Cheng, M.-M., & Gall, J. (2022). *Dual pyramid generative adversarial networks for semantic image synthesis*. In Proceedings of the 33rd British Machine Vision Conference (BMVC 2022) (p. 285). BMVA Press.
- [10] Henry, J., Natalie, T., & Madsen, D. (2021). *Pix2pix gan for image-to-image translation*. Research Gate Publication, 1-5.
- [11] Yu, F., Koltun, V., & Funkhouser, T. (2017). *Dilated residual networks*. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 472-480).
- [12] Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). *Gans trained by a two time-scale update rule converge to a local nash equilibrium*. Advances in neural information processing systems, 30.
- [13] Everingham, M., Van Gool, L., Williams, C. K., Winn, J., & Zisserman, A. (2010). *The pascal visual object classes (voc) challenge*. International journal of computer vision, 88, 303-338.



- **Generative Adversarial Networks** introduced by Ian Goodfellow et al. in 2014 [4].
- Consist of two neural networks: Generator and Discriminator, trained in opposition.
- Generator learns to *create realistic images* from noise or input data.
- Discriminator learns to *distinguish between real and fake* (generated) images.

- Introduces **SPADE** (**S**patially **A**daptive **D**enormalization) layers that inject semantic information directly into the generator.
- SPADE modulates the generator's activations based on the **input segmentation map at every layer**.
- Instead of concatenating segmentation maps with input noise, it uses them to dynamically modulate normalization parameters (scale & bias).
- Achieves state-of-the-art results on semantic image synthesis benchmarks like *Cityscapes* [7].





- A third pyramid is added to ***process the clear weather image*** as a separate modality.
- Purpose of the Third Pyramid:
 - Acts as a dedicated ***multi-scale encoder*** for extracting fine-grained appearance features from the clear-weather reference.
 - ***Complements*** the ***semantic*** structure with real-world texture, lighting, and detail cues.
 - Enables better modular separation of semantic and visual information.
 - Provides more detailed and ***context-aware conditioning*** throughout the generation process.

- Extended SPADE Conditioning:
 - Replaced single SPADE blocks with **dual SPADE** blocks at each decoder stage.
- Dual Normalization Streams: Each decoder feature map is processed through:
 - A SPADE block for structure (**semantic encoder pyramid**).
 - A SPADE block for appearance (**clear-weather encoder pyramid**).
- **Fusion Mechanism:** The outputs of the two SPADE blocks are fused using a learnable interpolation parameter (α).
- Final Output:

$$\text{Fused} = \alpha \cdot \text{SPADE}_{\text{seg}}(x) + (1 - \alpha) \cdot \text{SPADE}_{\text{clear}}(x),$$

$$\alpha \in [0, 1] \text{ via sigmoid}$$

$$f(x) = \frac{1}{1 + e^{-(x)}}$$

- Sigmoid as a smooth switch or soft gate.
- When $\alpha \approx 0$: rely on clear-weather features.
- When $\alpha \approx 1$: rely more on segmentation-based structure.
- The model learns when to trust each input.

$$\text{Fused} = \alpha \cdot \text{SPADE}_{\text{seg}}(x) + (1 - \alpha) \cdot \text{SPADE}_{\text{clear}}(x),$$

- Two new loss terms were added to the generator's objective:
 - **Feature Alignment Loss:**
 - Guide the generator to *mimic the internal representations* of real clear-weather scenes.
 - Computes an **L1 loss** between feature maps at selected layers.
 - **VGG Perceptual Loss:**
 - Encourage preservation of **texture, lighting, and surface properties** from the clear image.
 - Uses a *pretrained VGG-19* network to compare high-level features between the generated image and the clear reference image.

- Measure the difference between two sets of values — usually predictions and targets.
- Small error → the two feature maps are very similar.
- Large error → the generated image's features deviate from the reference.

$$\text{MSE}(f, f_{\text{ref}}) = \frac{1}{N} \sum_{i=1}^N (f_i - f_{\text{ref}, i})^2$$

- N is the number of elements (pixels × channels) in the feature map.
- f is the i -th value in the generated feature map.
- f_{ref} is the i -th value in the reference feature map.

- Based on **ResNet**, but modified for **dense prediction tasks** like semantic segmentation.
- Replaces standard downsampling layers with **dilated (atrous) convolutions** to expand the receptive field.
- Removes global average pooling and fully connected layers to preserve dense feature maps.
- Retains **residual connections** to enable deep, stable training.
- Combines local detail capture and global context.

- Used to translate from **clear to foggy** images **without semantic maps**
- Model training setup:
 - *100 epochs*
 - *Batch size = 2.*
 - *Using the Adam optimizer with a learning rate of 0.0002*
 - *Single NVIDIA GTX 1080 Ti GPU (11GB VRAM).*
- Input images were resized to 512x256.
- The original CycleGAN model was used without any architectural modifications.

- Evaluate the performance of **OASIS**, **DP-GAN** models on generating foggy images from ***segmentation maps only***.
- No clear-weather images were used to test generation quality from semantic labels.
- Identical preprocessing pipeline and dataset split for fair comparison.
- Input images were resized to 512x256.
- Models training setup:
 - ***400 epochs***
 - ***Batch size = 5***
 - ***Single NVIDIA RTX A5000 GPU (24 GB)***
- Enabled **VGG perceptual loss** for high-level texture guidance.
- Used original architecture without modifications.

- Used **Fréchet Inception Distance (FID)** to assess generated image quality[12].
- **FID** measures the distance between the feature distributions of real and generated images, using activations extracted from a pretrained Inception-v3 network.
- **Low FID $\Rightarrow$ better realism.**

- Measured by **segmentation performance** using **mean Intersection-over-Union (mIoU)** [13].
- First compute **IoU per class (IoU_k)**:

$$\text{IoU}_k = \frac{TP_k}{TP_k + FP_k + FN_k},$$

- Where:
 - TP_k: True Positives – correctly predicted pixels for class k.
 - FP_k: False Positives – predicted as k, but wrong.
 - FN_k: False Negatives – actually k, but predicted otherwise.
- Then compute **average** over all K classes.

$$\text{mIoU} = \frac{1}{K} \sum_{k=1}^K \text{IoU}_k.$$

- **Higher mIoU $\Rightarrow$ better semantic consistency.**

- Number of Parameters in Baseline Models vs Ours.

Method	FID
CycleGAN	~22 million
OASIS	~45 million
DP-GAN	~50 million
3P-WeatherGAN (Ours)	~60 million

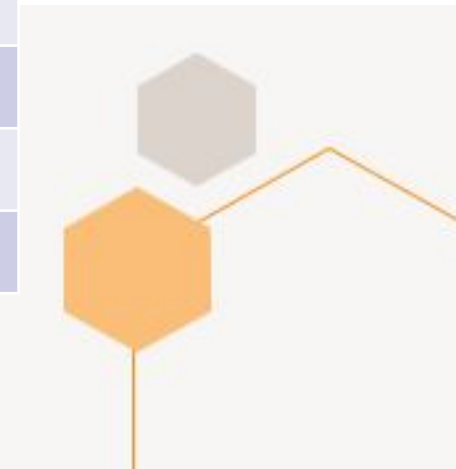
Table: Quantitative Comparision — Number of Parameters



- Evaluation of **segmentation task** on images generated by CycleGAN (**no segmentation input**) versus our proposed method (**segmentation-guided**) based on mean Intersection-over-Union (**mIoU**) — **higher is better**.
- The segmentation model was trained on **varying ratios** of real foggy images and GAN-generated images.
- Scores were calculated on ACDC Fog Validation Set with matching annotations (50 Images).

Training Dataset		mIoU
Real Fog Only		36.3
Real Fog + 200 synthetic images	CycleGAN	38.12
	Ours	41.42
Real Fog + 400 synthetic images	CycleGAN	42.07
	Ours	42.05

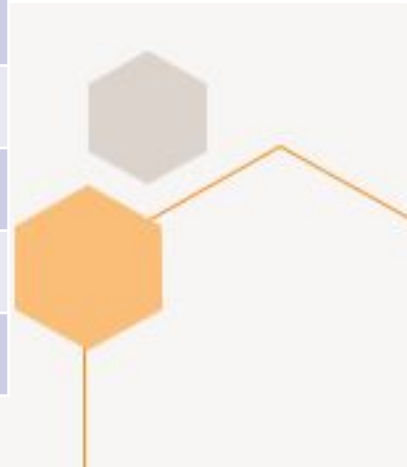
Table: Quantitative Comparison — mIoU results



- Evaluation of **segmentation task** on images generated by CycleGAN (**no segmentation input**) versus our proposed method (**segmentation-guided**) based on mean Intersection-over-Union (**mIoU**) — **higher is better**.
- The segmentation model was trained on **varying ratios** of real foggy images and GAN-generated images.
- Scores were calculated on ACDC Fog Validation Set with matching annotations (50 Images).

Training Dataset		mIoU
Fog Only		36.3
Fog + 200 synthetic images	CycleGAN	38.12
	Ours	41.42
Fog + 400 synthetic images	CycleGAN	42.07
	Ours	42.05
Fog + 800 synthetic images	CycleGAN	45.57
	Ours	43.61

Table: Quantitative Comparison — mIoU results



- Other clear weather image incorporation techniques.
- **Channel Fusion:** Direct concatenation of clear weather images on channel dimension 34 (segmentation classes) + 3 (RGB clear image) channel.
- **Three - Pyramid Fusion:**
 - **Single SPADE :** Concatenate both Pyramids at the Resnet SPADE blocks inputs.
 - **Dual Parallel SPADE:** Have 2 separate SPADE blocks for each pyramid, the resulting intermediate feature maps are fused directly.
 - **Dual Sequential SPADE:** Have 2 separate SPADE blocks for each pyramid, then apply SPADE normalization sequentially i.e input will pass through Segmentation SPADE then Clear Weather SPADE.

Method	FID
Pyramid Fusion - Dual Sequential SPADE	111.52
Pyramid Fusion - Single SPADE	108.21
Channel Fusion	99.07
Pyramid Fusion - Dual Parallel SPADE	98.01

Table: Quantitative Ablation Study — FID results